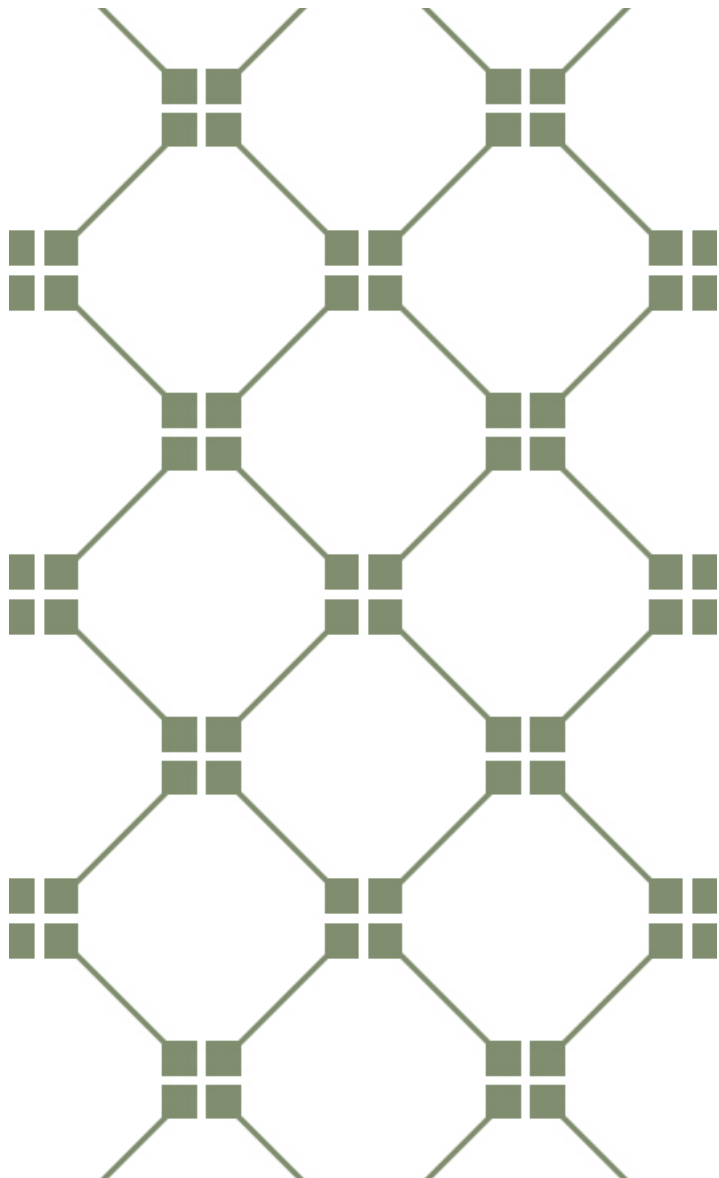


WARM-UP

- ❖ What are the charges (oxidation numbers) for the following ions: Be, N, and Br
- ❖ Write the isotope notation for gallium ion with 40 neutrons
- ❖ Complete an electron configuration and dot diagram for nitrogen *ion*.

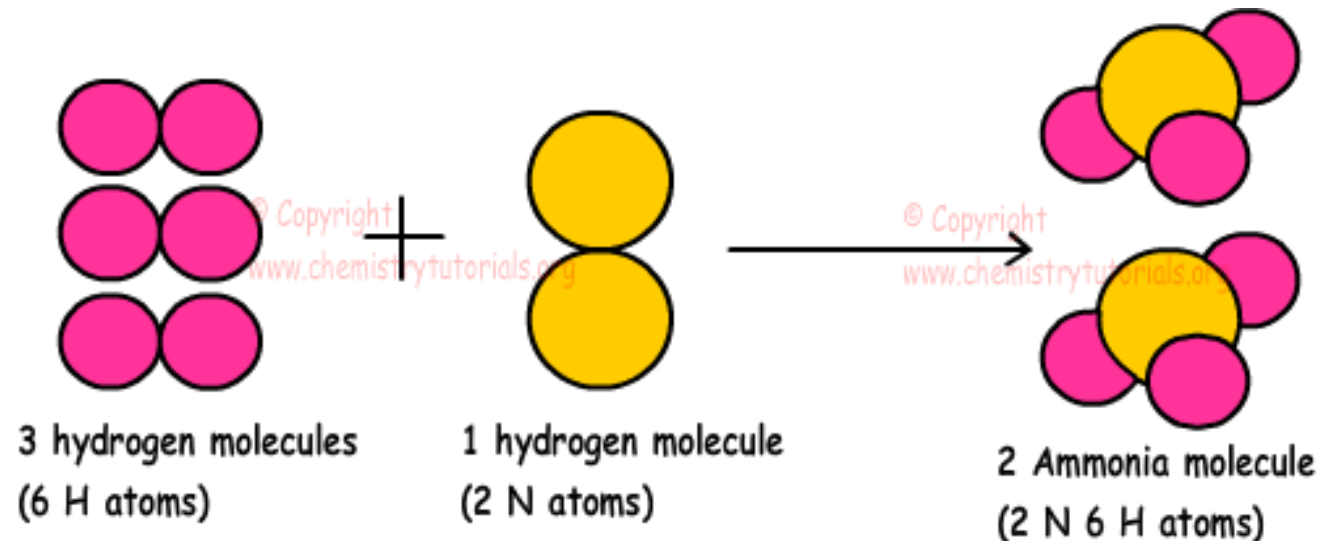


UNIT 3- BONDING

Hoffman Honors Chemistry

WHAT IS A CHEMICAL BOND?

It is the mutual electrical attraction between nuclei and valence electrons of different atoms



TO BECOME
STABLE!!



Key: 8
valence
electrons

WHY DO ATOMS BOND?

THREE TYPES OF BONDS

1. Ionic – electrostatic force of attraction between (+) and (-) ions
2. Covalent – sharing of electron pairs between nonmetal atoms
 - 4 Sub-types: Polar, Nonpolar, Coordinate, Network
3. Metallic – attraction between metal cations and outer mobile electrons

Type of Bond	Representative Unit	What bonds? (M, NM, MD)	How do they bond? (hint: electrons)	Example
Ionic	Formula unit	Metal – Nonmetal	Transfer electrons	CaBr ₂ BaSO ₄
Covalent	Molecule	Nonmetals	Share electrons	NH ₃ H ₂ S H ₃ PO ₄
Metallic	Atom	Metals	Free delocalized electrons	platinum Silver

MAIN QUESTIONS FOR EACH BOND TYPE

What atoms combine to make the bond?

How do the atoms combine together?

What properties result from the bond type created?

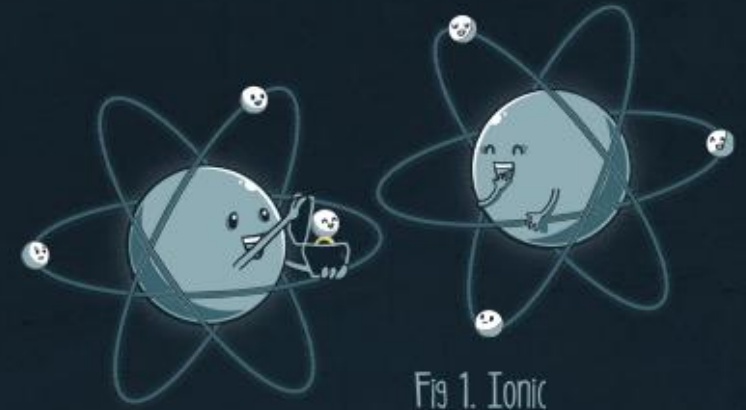


Fig 1. Ionic

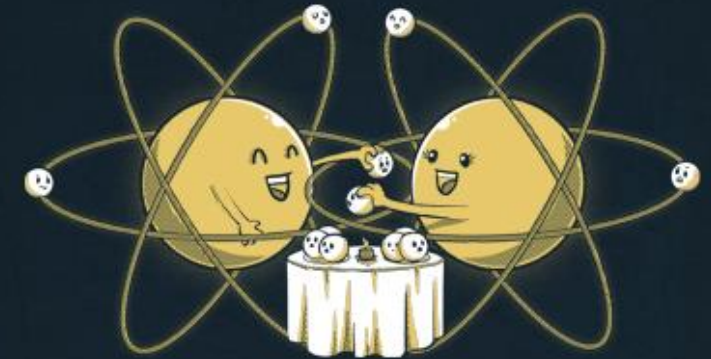


Fig 2. Covalent

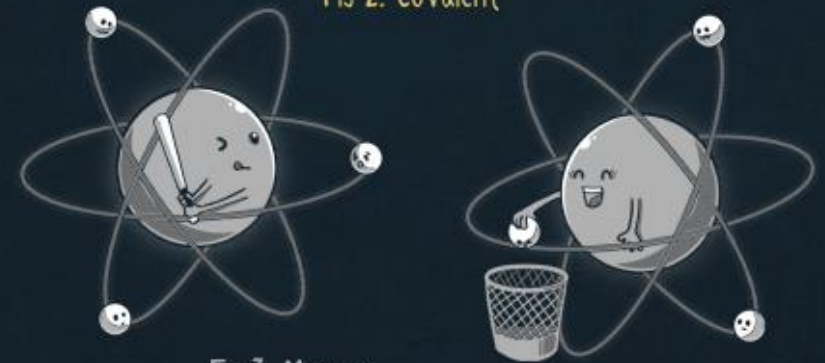


Fig 3. Metallic

IONIC BONDING



Occurs between **METALS** and **NONMETALS**

Valence electrons are transferred from the metal to the nonmetal.

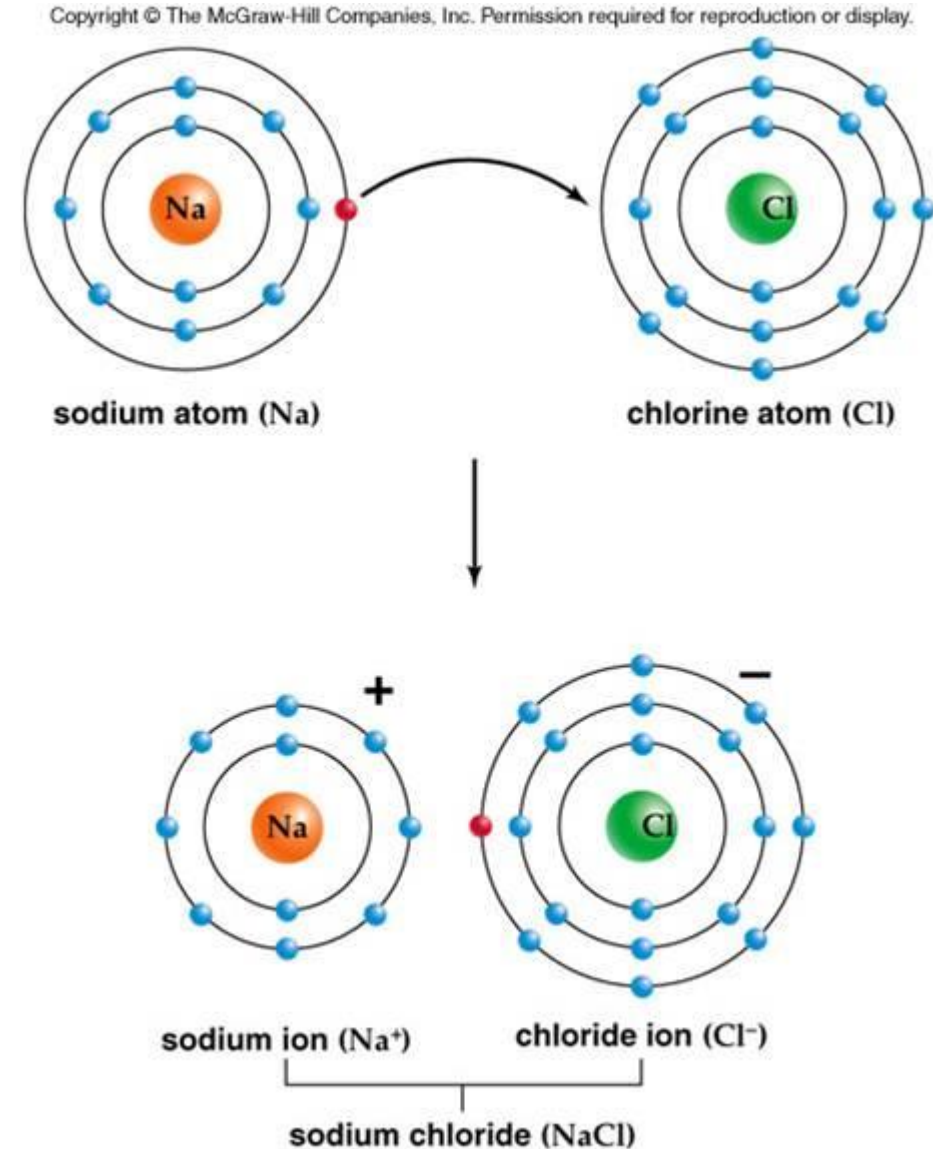
Results in a (+) metal cation and a (-) nonmetal anion that are attracted to each other electrostatically

IONIC BONDING:

Na metal loses its one valence e-

Cl nonmetal gains the valence e-
from the Na metal atom

Creates Na^+ and Cl^-



KEY THINGS TO REMEMBER

METALS

Metals are located to the left of the staircase on the periodic table

Metal atoms lose valence electrons

Become positively charged

Called cations

NONMETALS

Nonmetals are located to the right of the staircase on the periodic table

Nonmetal atoms gain valence electrons

Become negatively charged

Called anions

1 IA												18 VIIIA					
1 H Hydrogen 1.008	2 IIA											13 IIIA	14 IVA	15 VA	16 VIA	17 VIIA	2 He Helium 4.003
3 Li Lithium 6.941	4 Be Beryllium 9.012											5 B Boron 10.81	6 C Carbon 12.01	7 N Nitrogen 14.01	8 O Oxygen 16.00	9 F Fluorine 19.00	10 Ne Neon 20.18
11 Na Sodium 22.99	12 Mg Magnesium 24.31	3 IIIB	4 IVB	5 VB	6 VIB	7 VIIB	8 VIIIB	9 VIIIB	10 VIIIB	11 IB	12 IIB	13 Al Aluminum 26.98	14 Si Silicon 28.09	15 P Phosphorus 30.97	16 S Sulfur 32.07	17 Cl Chlorine 35.45	18 Ar Argon 39.95
19 K Potassium 39.10	20 Ca Calcium 40.08	21 Sc Scandium 44.96	22 Ti Titanium 47.88	23 V Vanadium 50.94	24 Cr Chromium 51.99	25 Mn Manganese 54.94	26 Fe Iron 55.85	27 Co Cobalt 58.93	28 Ni Nickel 58.69	29 Cu Copper 63.55	30 Zn Zinc 65.39	31 Ga Gallium 69.72	32 Ge Germanium 72.61	33 As Arsenic 74.92	34 Se Selenium 78.96	35 Br Bromine 79.90	36 Kr Krypton 83.80
37 Rb Rubidium 85.47	38 Sr Strontium 87.62	39 Y Yttrium 88.91	40 Zr Zirconium 91.22	41 Nb Niobium 92.91	42 Mo Molybdenum 95.94	43 Tc Technetium (98)	44 Ru Ruthenium 101.07	45 Rh Rhodium 102.91	46 Pd Palladium 106.42	47 Ag Silver 107.87	48 Cd Cadmium 112.41	49 In Indium 114.82	50 Sn Tin 118.71	51 Sb Antimony 121.76	52 Te Tellurium 127.60	53 I Iodine 126.90	54 Xe Xenon 131.29
55 Cs Cesium 132.91	56 Ba Barium 137.38	57 La Lanthanum 138.91	72 Hf Hafnium 178.49	73 Ta Tantalum 180.95	74 W Tungsten 183.84	75 Re Rhenium 186.21	76 Os Osmium 190.23	77 Ir Iridium 192.22	78 Pt Platinum 195.08	79 Au Gold 196.97	80 Hg Mercury 200.59	81 Tl Thallium 204.38	82 Pb Lead 207.20	83 Bi Bismuth 208.98	84 Po Polonium (209)	85 At Astatine (210)	86 Rn Radon (222)
87 Fr Francium (223)	88 Ra Radium (226)	89 Ac Actinium (227)	104 Rf Rutherfordium (261)	105 Db Dubnium (262)	106 Sg Seaborgium (263)	107 Bh Bohrium (264)	108 Hs Hassium (269)	109 Mt Meitnerium (268)	110 Ds Darmstadtium (271)	111 Rg Roentgenium (272)	112 Cn Copernicium (285)						

58 Ce Cerium 140.12	59 Pr Praseodymium 140.91	60 Nd Neodymium 144.24	61 Pm Promethium (145)	62 Sm Samarium 150.36	63 Eu Europium 151.96	64 Gd Gadolinium 157.25	65 Tb Terbium 158.93	66 Dy Dysprosium 162.50	67 Ho Holmium 164.93	68 Er Erbium 167.26	69 Tm Thulium 168.93	70 Yb Ytterbium 173.04	71 Lu Lutetium 174.97
90 Th Thorium 232.04	91 Pa Protactinium 231.04	92 U Uranium 238.04	93 Np Neptunium (237)	94 Pu Plutonium (244)	95 Am Americium (243)	96 Cm Curium (247)	97 Bk Berkelium (247)	98 Cf Californium (251)	99 Es Einsteinium (252)	100 Fm Fermium (257)	101 Md Mendelevium (258)	102 No Nobelium (254)	103 Lr Lawrencium (262)

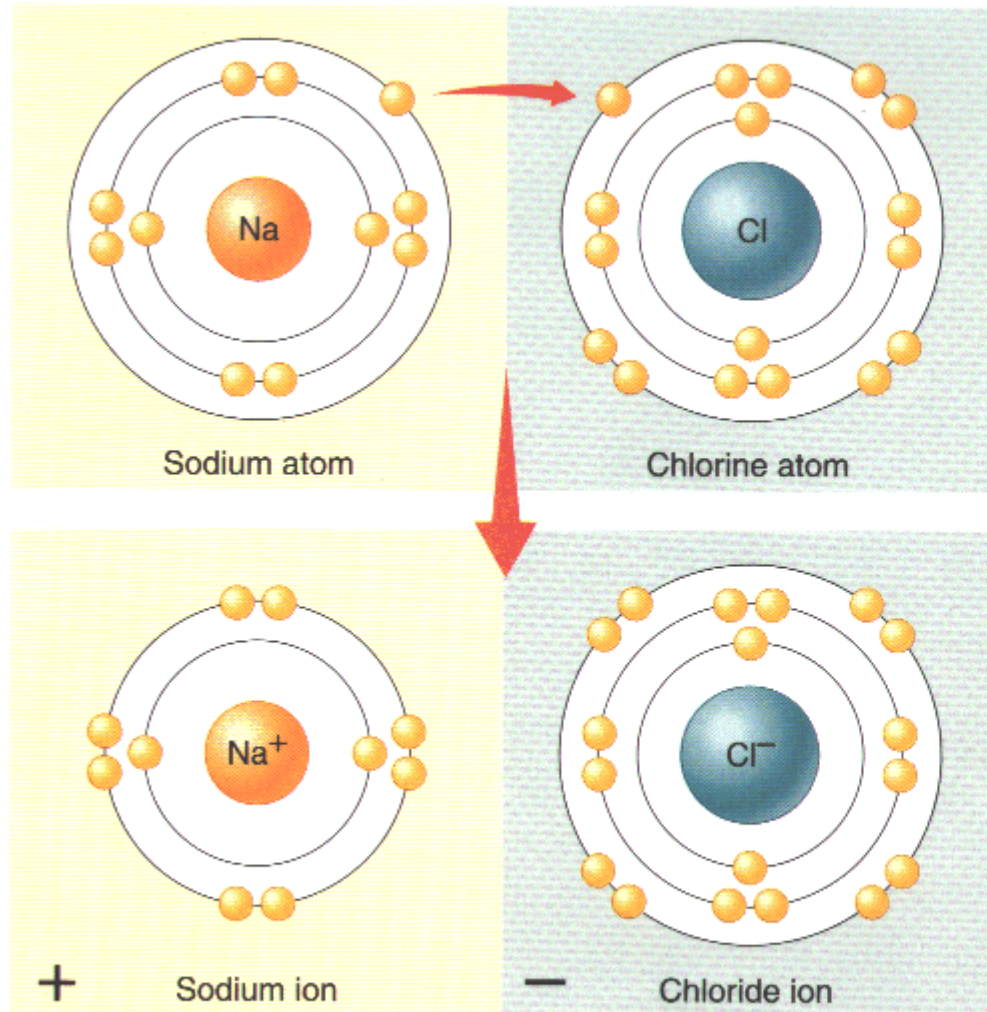
KEY VOCABULARY

Ionic bonding results in the formation of **ionic compounds**: compound formed by the electrostatic force of attraction between (+) and (-) ions

Formula unit: simplest combining ratio of ions in a compound; does not exist independently

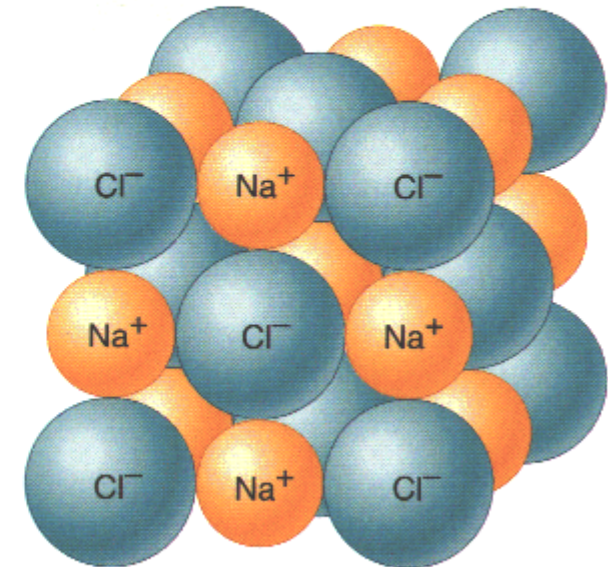
These ions align with each other to create a **crystal lattice**: 3D arrangement of ions

This creates **lattice energy**: energy released when one mole of ionic crystalline compound is formed from gaseous ions



(a)

Crystal Lattice: each (+) ion is completely surrounded by (-) ions and each (-) ion is completely surrounded by (+) ions



NaCl crystal

(b)

PROPERTIES OF IONIC COMPOUNDS

High melting points and boiling points

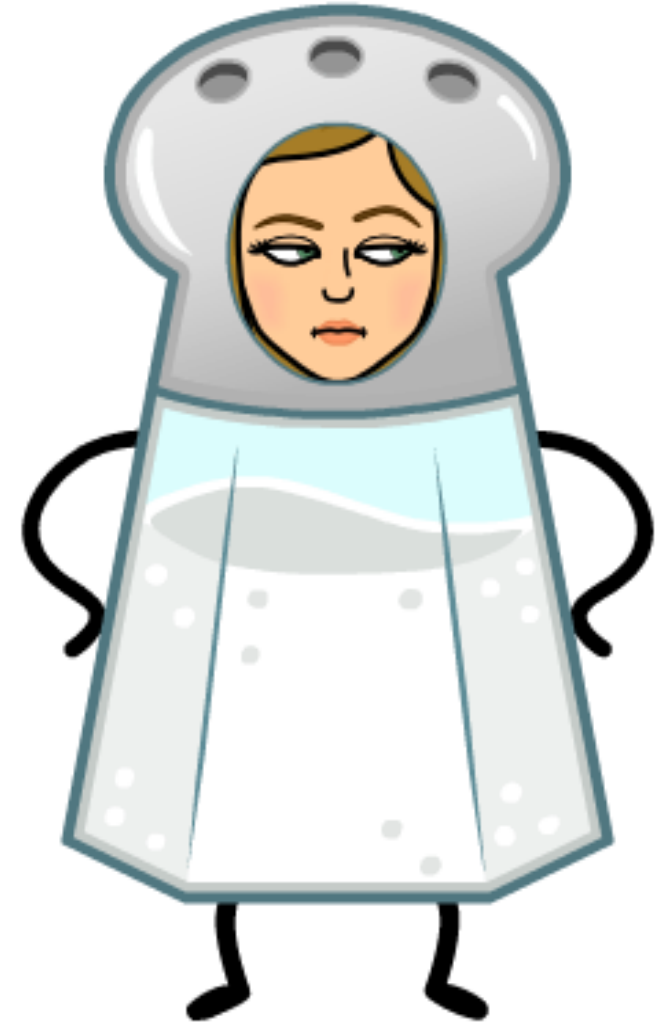
- Because they are a strong bond and require a lot of energy to break

Soluble in water

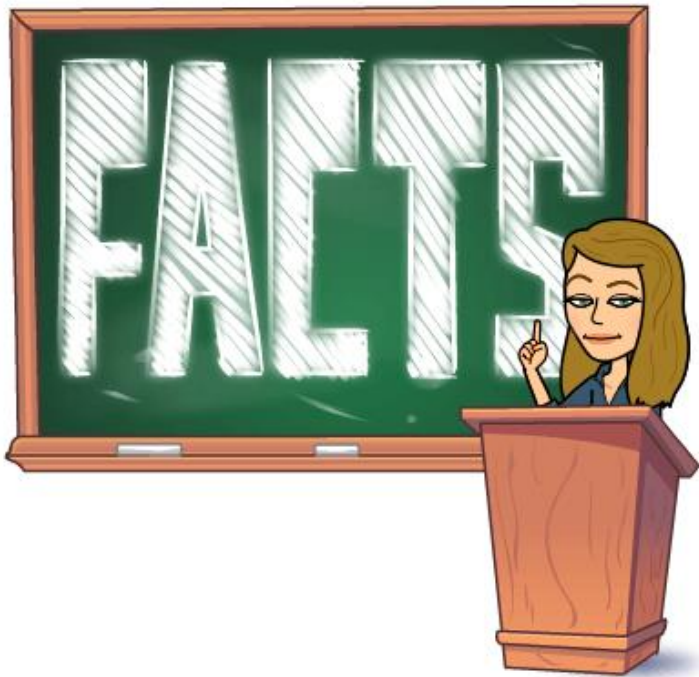
- Because the charged ions are more strongly attracted to water than to each other

Hard crystalline solids but can fracture

- If crystal structure is hit and ions shift so that like charges are next to each other, then the resulting repulsive forces will cause the crystal structure to break/fracture



PROPERTIES OF IONIC COMPOUNDS



Good conductors in liquid or dissolved states (electrolytes)

- Because the ions are free to move

Poor conductors in solid state

- Because the ions are NOT free to move

WRITING IONIC FORMULAS

1. Write the metal/cation first, nonmetal/anion 2nd
2. Write the oxidation #'s (charges) as superscripts for the metal and nonmetal
3. Criss-Cross the charges and make the numbers subscripts.
4. Reduce if necessary (do not reduce the subscript with a P.A ion)
5. Polyatomic ions stay in parenthesis, unless no number comes after it.

PRACTICE- WRITE THE IONIC COMPOUND

Remember: Overall charge must be neutral!

Lithium and fluorine

Calcium and chlorine

Sodium and sulfur

Magnesium and phosphorus

Beryllium and oxygen

DRAWING DOT DIAGRAMS FOR IONIC STRUCTURES

Arrows show transfer on electrons from metal to nonmetal.

LET'S PRACTICE! LEWIS DOT STRUCTURES

Complete Lewis dot structures for the following

Calcium and chlorine

Sodium and sulfur

Beryllium and oxygen

WARM-UP

Write the formula and Lewis dot structure for the following:

- Fe (+3) and Cl
- Be and F
- Pb (+4) and O

What is the formula for a compound of magnesium and nitrate?

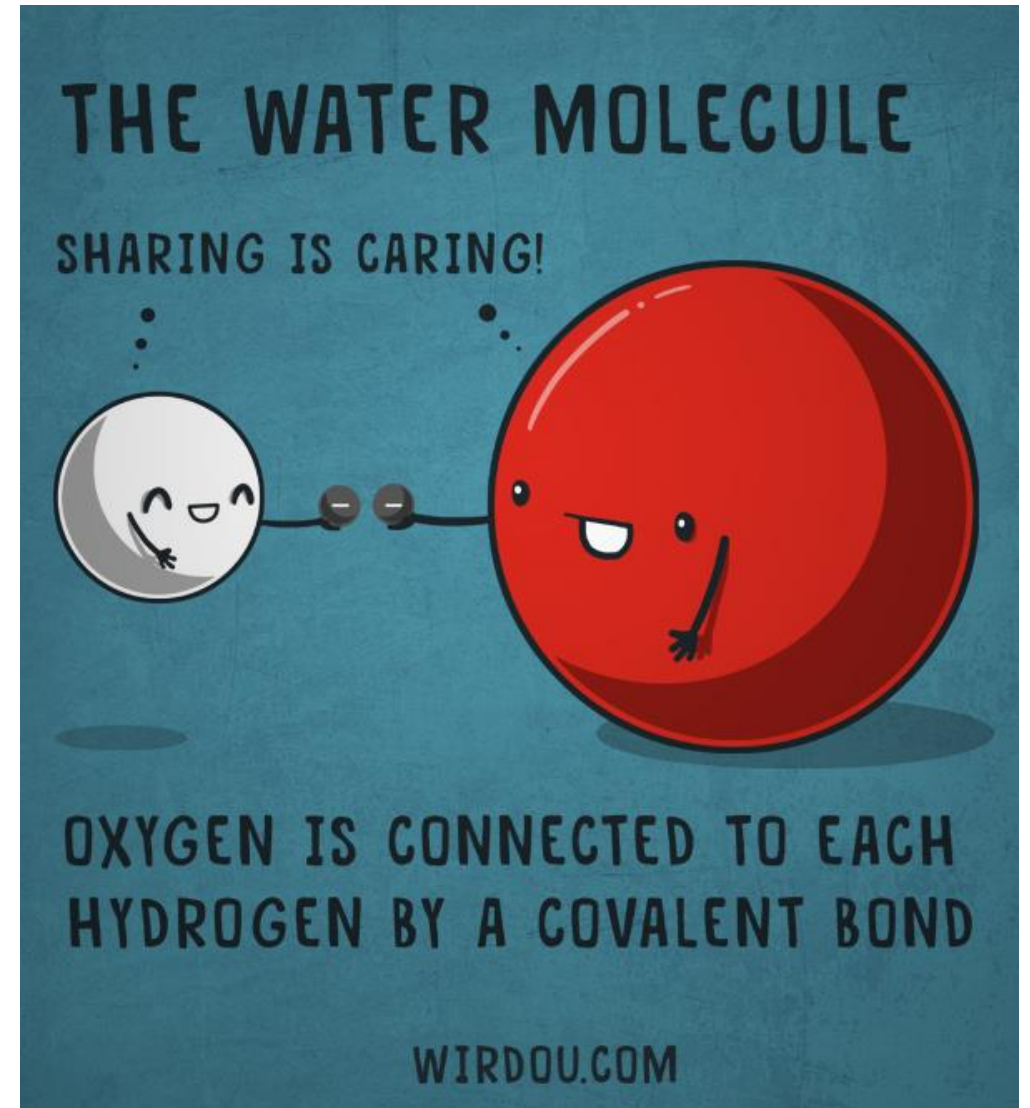
2:00

COVALENT BONDING

Occurs between **NONMETALS**

Nonmetals have high electronegativities and want to gain more electrons. They cannot lose valence electrons. In order to bond, nonmetals must then **SHARE** valence electrons.

The atoms share enough electrons to obtain 8 valence electrons (including the shared electrons)



LET'S TRY IT!

Is it ionic or
covalent?



1																	18
1 H 1.0079	2											13	14	15	16	17	2 He 4.0026
3 Li 6.941	4 Be 9.0122											5 B 10.811	6 C 12.011	7 N 14.007	8 O 15.999	9 F 18.998	10 Ne 20.180
11 Na 22.990	12 Mg 24.305	3	4	5	6	7	8	9	10	11	12	13 Al 26.982	14 Si 28.086	15 P 30.974	16 S 32.065	17 Cl 35.453	18 Ar 39.948
19 K 39.098	20 Ca 40.078	21 Sc 44.956	22 Ti 47.867	23 V 50.942	24 Cr 51.996	25 Mn 54.938	26 Fe 55.845	27 Co 58.933	28 Ni 58.693	29 Cu 63.546	30 Zn 65.409	31 Ga 69.723	32 Ge 72.64	33 As 74.922	34 Se 78.96	35 Br 79.904	36 Kr 83.798
37 Rb 85.468	38 Sr 87.62	39 Y 88.906	40 Zr 91.224	41 Nb 92.906	42 Mo 95.94	43 Tc (98)	44 Ru 101.07	45 Rh 102.91	46 Pd 106.42	47 Ag 107.87	48 Cd 112.41	49 In 114.82	50 Sn 118.71	51 Sb 121.76	52 Te 127.60	53 I 126.90	54 Xe 131.29
55 Cs 132.91	56 Ba 137.33	57-71 *	72 Hf 178.49	73 Ta 180.95	74 W 183.84	75 Re 186.21	76 Os 190.23	77 Ir 192.22	78 Pt 195.08	79 Au 196.97	80 Hg 200.59	81 Tl 204.38	82 Pb 207.2	83 Bi 208.98	84 Po (209)	85 At (210)	86 Rn (222)
87 Fr (223)	88 Ra (226)	89-103 #	104 Rf (261)	105 Db (262)	106 Sg (266)	107 Bh (264)	108 Hs (277)	109 Mt (268)	110 Ds (281)	111 Rg (272)	112 Uub (285)	113 Uut (284)	114 Uuq (289)	115 Uup (288)			

* Lanthanide
series

57 La 138.91	58 Ce 140.12	59 Pr 140.91	60 Nd 144.24	61 Pm (145)	62 Sm 150.36	63 Eu 151.96	64 Gd 157.25	65 Tb 158.93	66 Dy 162.50	67 Ho 164.93	68 Er 167.26	69 Tm 168.93	70 Yb 173.04	71 Lu 174.97
---------------------------	---------------------------	---------------------------	---------------------------	--------------------------	---------------------------	---------------------------	---------------------------	---------------------------	---------------------------	---------------------------	---------------------------	---------------------------	---------------------------	---------------------------

Actinide
series

89 Ac (227)	90 Th 232.04	91 Pa 231.04	92 U 238.03	93 Np (237)	94 Pu (244)	95 Am (243)	96 Cm (247)	97 Bk (247)	98 Cf (251)	99 Es (252)	100 Fm (257)	101 Md (258)	102 No (259)	103 Lr (262)
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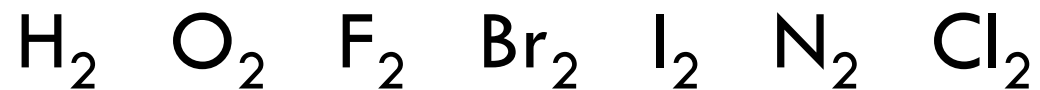
KEY VOCABULARY

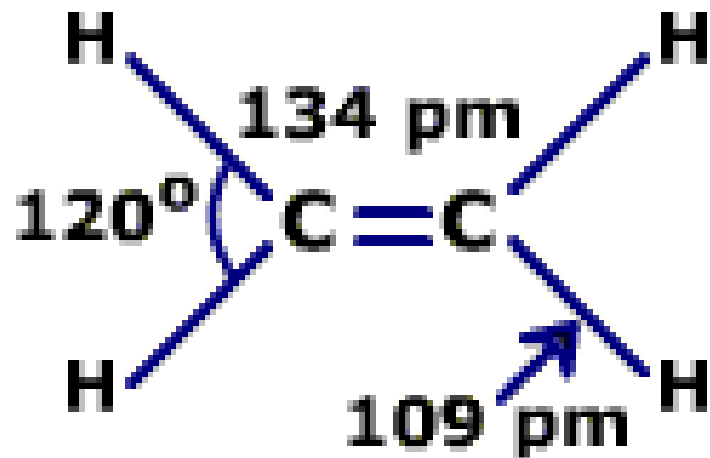
Molecular Compound: neutral compound consisting of nonmetals covalently bonded

Molecule: smallest representative unit of a molecular compound; CAN exist independently

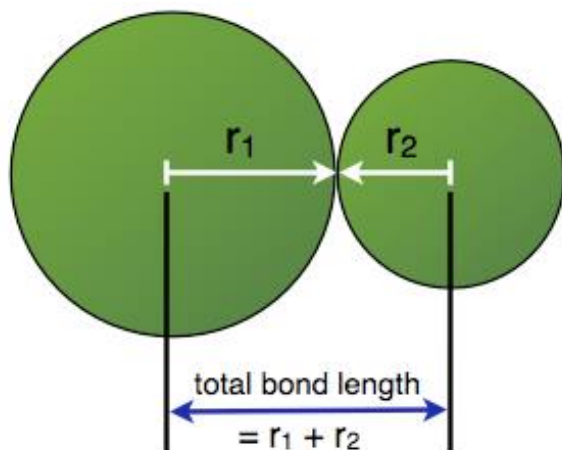
Diatomic Molecules: molecules consisting of two atoms of the same element covalently bonded

■ “HOFBrINCl”





(c)



WHEN MOLECULES FORM THE RESULTING BOND HAS A LENGTH, ENERGY AND ANGLE ASSOCIATED WITH IT

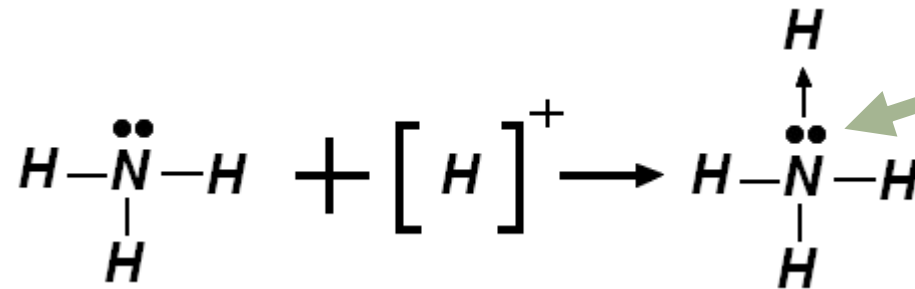
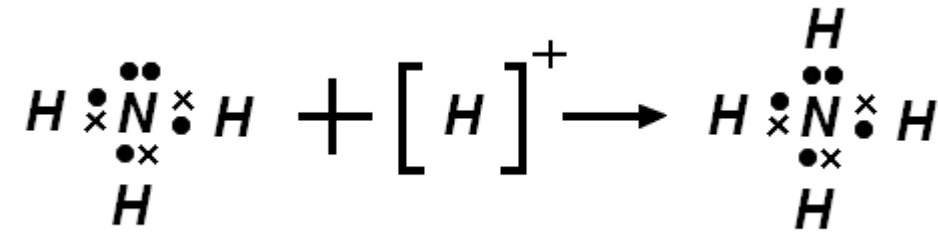
Bond Length: average distance between nuclei of two bonded nonmetal atoms (sum of atomic radii)

Bond Angle: angle between two bonds in a molecule

Bond Energy: energy needed to break a bond and form neutral atoms

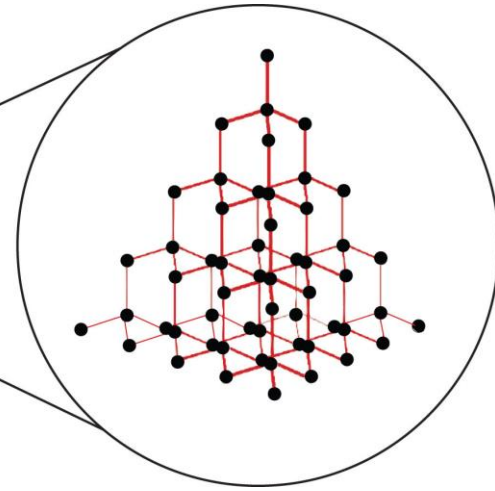
COORDINATE COVALENT BOND

Definition: Results when both electrons to be shared in the covalent bond come from one atom



Coordinate
Covalent bond
will form here

NETWORK COVALENT BOND



Definition: 3D network of covalently bonded atoms that extend in all directions to make large crystalline structures

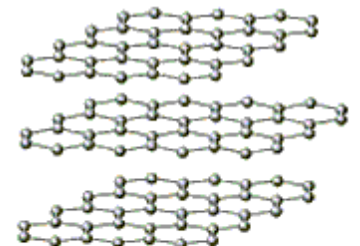
Examples: Diamond C(s), Graphite C(s), Sand SiO₂(s), Silicon Carbide SiC(s)

Properties:

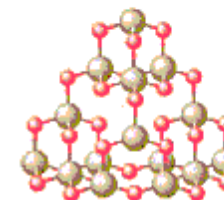
- Poor conductors
- Very hard solids
- Very high melting and boiling points
- Insoluble in water



Diamond



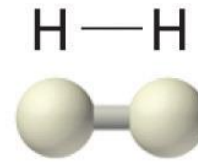
Graphite



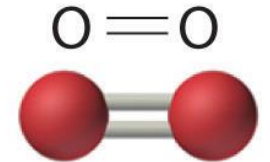
Silica

Single Bond

- 2 electrons shared
- Represented as a single line between atoms



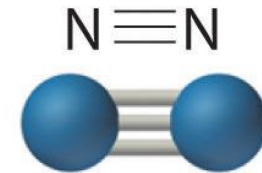
Single bond



Double bond

Double Bond

- 4 electrons shared
- Represented as a double line between atoms



Triple bond

Triple Bond

- 6 electrons shared
- Represented as a triple line between atoms

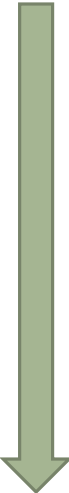
Single Bond

Weakest bond, longest bond, least amount of energy

Double Bond

Triple Bond

Strongest bond, shortest bond, most amount of energy



Bond strength increases as the amount of e- shared increases resulting in shorter bonds with greater energy

POLYATOMIC IONS

Covalent compounds that have a charge.

They can then be involved in ionic compounds!

You will need to memorize these!

Reference Table

Polyatomic Ions	
NH_4^+	Ammonium
BrO_3^-	Bromate
CN^-	Cyanide
$\text{C}_2\text{H}_3\text{O}_2^-$ (CH_3COO^-)	Acetate
ClO_4^-	Perchlorate
ClO_3^-	Chlorate
ClO_2^-	Chlorite
ClO^-	Hypochlorite
IO_3^-	Iodate
MnO_4^-	Permanganate
NO_3^-	Nitrate
NO_2^-	Nitrite
OH^-	Hydroxide
HCO_3^-	Hydrogen carbonate
HSO_4^-	Hydrogen sulfate
SCN^-	Thiocyanate
CO_3^{2-}	Carbonate
$\text{Cr}_2\text{O}_7^{2-}$	Dichromate
CrO_4^{2-}	Chromate
SO_4^{2-}	Sulfate
SO_3^{2-}	Sulfite
PO_4^{3-}	Phosphate

IONIC COMPOUNDS W/ POLYATOMIC IONS

Sodium and carbonate (CO_3^{2-})

Calcium and hydroxide (OH^-)

PROPERTIES OF MOLECULES

Gases or dull, brittle solids

Poor conductors of heat and electricity

- Because e⁻ are stuck in the bonds and cannot move

Low melting and boiling points

- Because these are weak bonds and easy to break requiring little energy



PROPERTIES OF MOLECULES

Solubility depends on

“like dissolve like”

- Polar substances will dissolve in other polar substances
- Nonpolar substances will dissolve in other nonpolar substances
- Polar and Nonpolar substances will NOT dissolve in each other



LEWIS DOT STRUCTURES — COVALENT COMPOUNDS

Usually: least electronegative compound goes in the middle

Must follow the Octet Rule: atoms tend to gain, lose, or share valence electrons so that they have 8 valence electrons

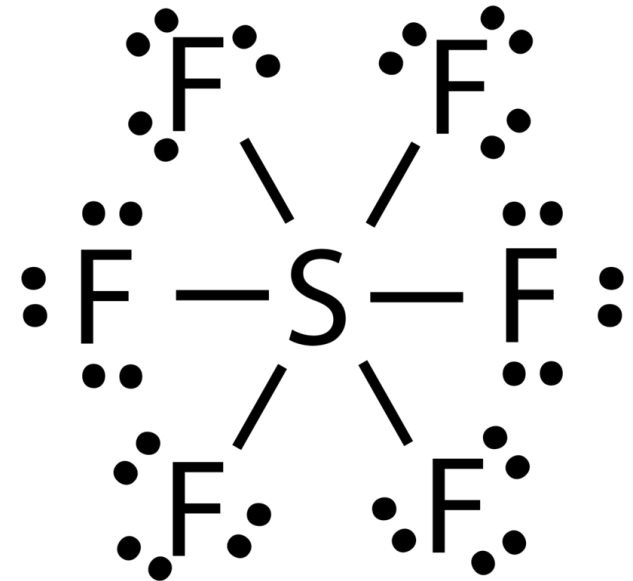
- **Exceptions:** H (2 e-), Be (4 e-), B (6 e-)

Before and after: count total number of valence electrons

OCTET RULE EXCEPTIONS

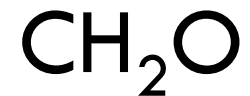
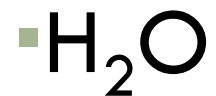
Expanded octet $\rightarrow$ more than 8 valence e^- (ex: **S**, **P**, **Xe**)

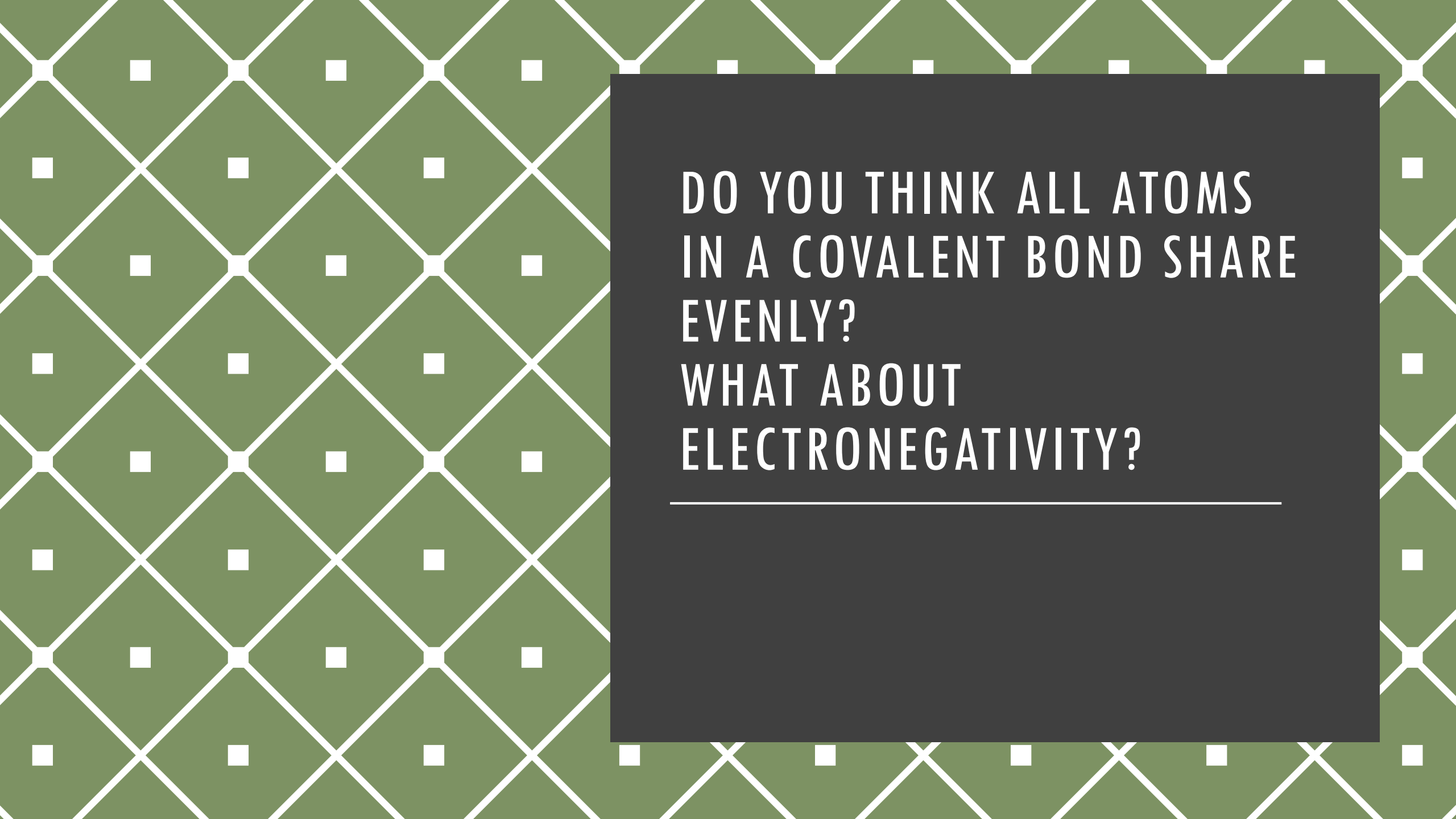
Radicals $\rightarrow$ odd # of valence



LEWIS DOT STRUCTURES PRACTICE

Let's try it!





DO YOU THINK ALL ATOMS
IN A COVALENT BOND SHARE
EVENLY?
WHAT ABOUT
ELECTRONEGATIVITY?

POLARITY

The valence electrons can be shared equally or unequally

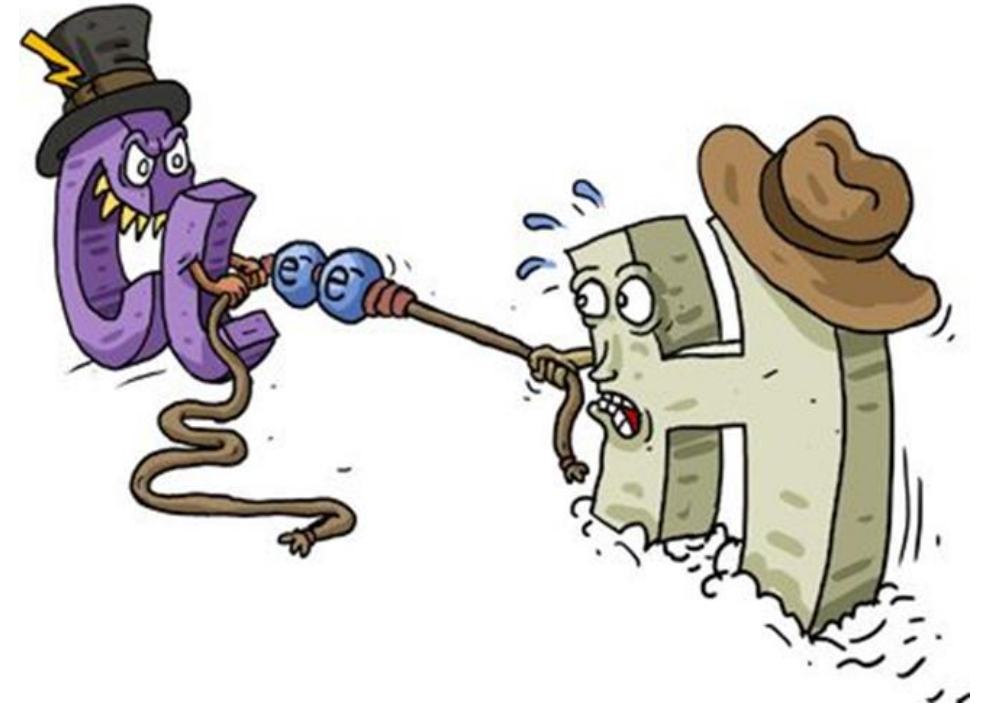
Polar covalent bond – valence electrons are shared unequally

- ex: H_2O , NH_3

Nonpolar covalent bond – valence electrons are shared equally

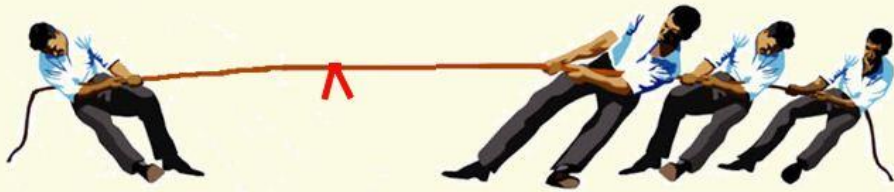
- Ex: F_2 , I_2

Polar Covalent Bond



POLAR VS. NONPOLAR COVALENT BONDS

Polar Covalent Bond



A polar covalent bond is a bond formed when the shared pair of electrons in a covalent bond are not shared equally.

This is due to different elements electronegativities.

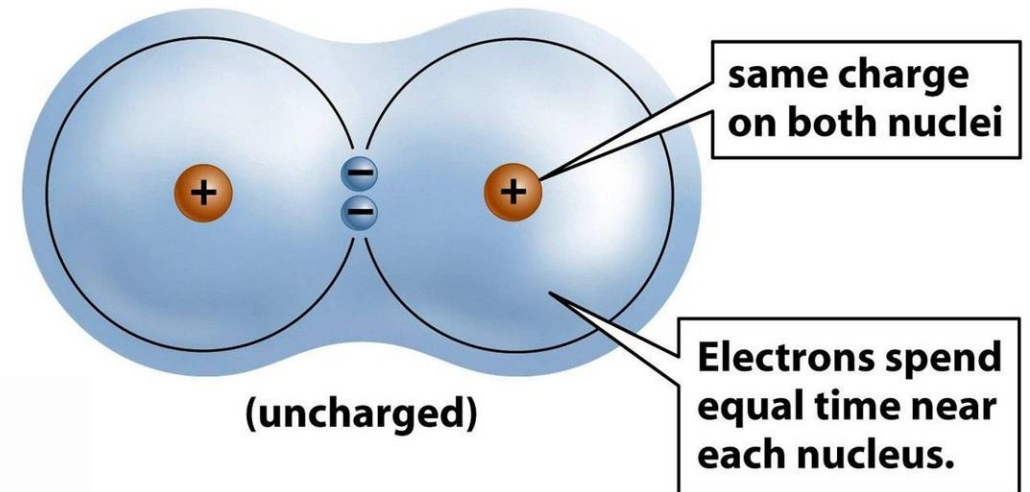


Less electron density
"Positive" end of the molecule: δ^+

More electron density
"Negative" end of the molecule: δ^-

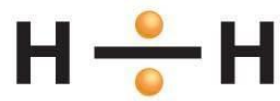
Nonpolar covalent bonding

Hydrogen
(H_2 or $H-H$)



ology: Life on Earth, 8/e
Prentice Hall, Inc.

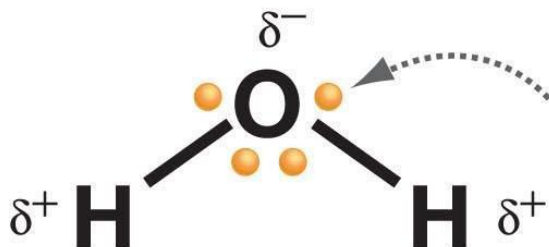
(a) Nonpolar covalent bond in hydrogen molecule



Electrons are shown to be superimposed on the bond to indicate that they are halfway between the two atoms, shared equally

Nonpolar covalent bonds occur between nonmetal atoms of same/similar electronegativities.

(b) Polar covalent bonds in water molecule



Electrons are not shared equally (O is more electronegative than H), so partial charges exist on the O and H atoms

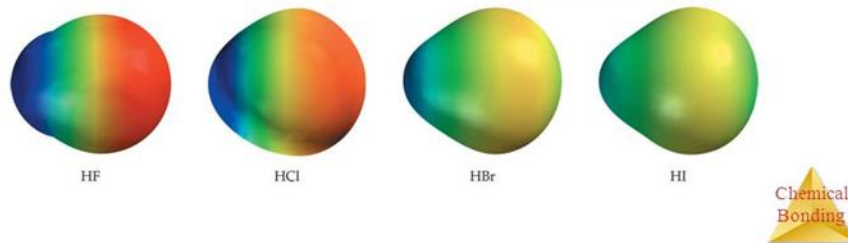
Polar covalent bonds occur between nonmetal atoms with different electronegativities

DETERMINING BOND TYPE/CHARACTER FROM ELECTRONEGATIVITY

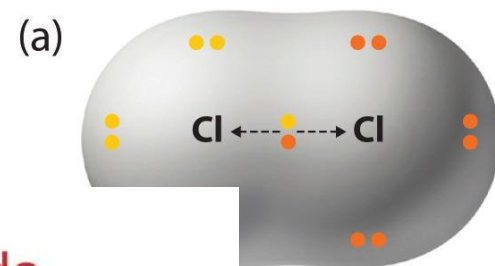
Polar Covalent Bonds

Compound	Bond Length (Å)	Electronegativity Difference	Dipole Moment (D)
HF	0.92	1.9	1.82
HCl	1.27	0.9	1.08
HBr	1.41	0.7	0.82
HI	1.61	0.4	0.44

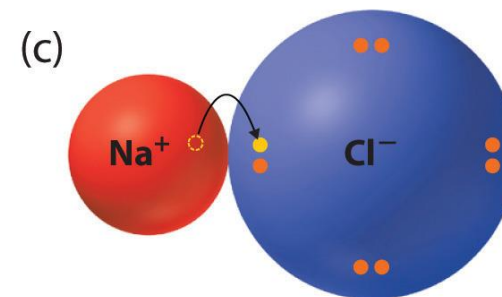
The greater the difference in electronegativity, the more polar is the bond.



bonding electrons shared unequally between two atoms. Partial charges on atoms.



valent bond
ons shared
en two atoms.
atoms.



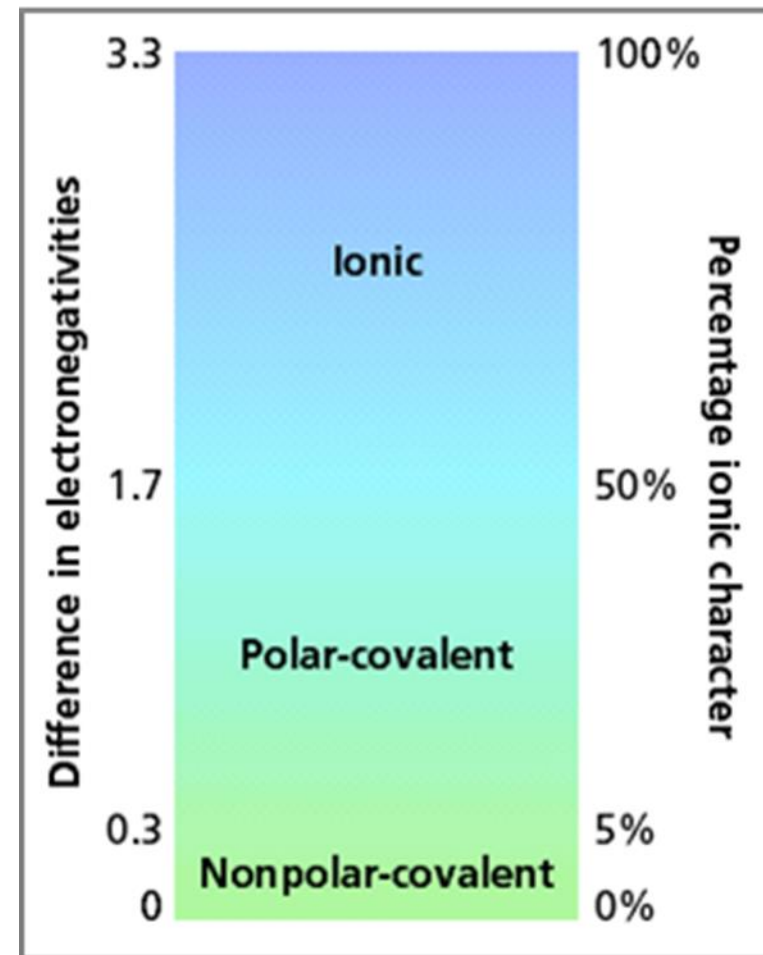
Ionic bond

Complete transfer of one or more valence electrons.
Full charges on resulting ions.

PREDICTING BOND TYPE USING ELECTRONEGATIVITIES

Find the difference in electronegativity

- 0 – 0.3 Non-polar covalent
- >0.3 – 1.7 Polar-covalent
- >1.7 – 3.3 Ionic



H 2.1																	He —		
Li 1.0	Be 1.5													B 2.0	C 2.5	N 3.0	O 3.5	F 4.0	Ne —
Na 0.9	Mg 1.2													Al 1.5	Si 1.8	P 2.1	S 2.5	Cl 3.0	Ar —
K 0.8	Ca 1.0	Sc 1.3	Ti 1.5	V 1.6	Cr 1.6	Mn 1.5	Fe 1.8	Co 1.8	Ni 1.8	Cu 1.9	Zn 1.6	Ga 1.6	Ge 1.8	As 2.0	Se 2.4	Br 2.8	Kr —		
Rb 0.8	Sr 1.0	Y 1.2	Zr 1.4	Nb 1.6	Mo 1.8	Tc 1.9	Ru 2.2	Rh 2.2	Pd 2.2	Ag 1.9	Cd 1.7	In 1.7	Sn 1.8	Sb 1.9	Te 2.1	I 2.5	Xe —		
Cs 0.7	Ba 0.9	La-Lu 1.1-1.2	Hf 1.3	Ta 1.5	W 1.7	Re 1.9	Os 2.2	Ir 2.2	Pt 2.2	Au 2.4	Hg 1.9	Tl 1.8	Pb 1.8	Bi 1.9	Po 2.0	At 2.2	Rn —		

PRACTICE PROBLEMS

H – S (2.1, 2.5)

Cs – S (0.7, 2.5)

S – Cl (2.5, 3.0)

O – O (3.5, 3.5)

PRACTICE PROBLEMS

H – S (2.1, 2.5)

Polar Covalent bond

S – Cl (2.5, 3.0)

Polar Covalent bond

Cs – S (0.7, 2.5)

Ionic bond

O – O (3.5, 3.5)

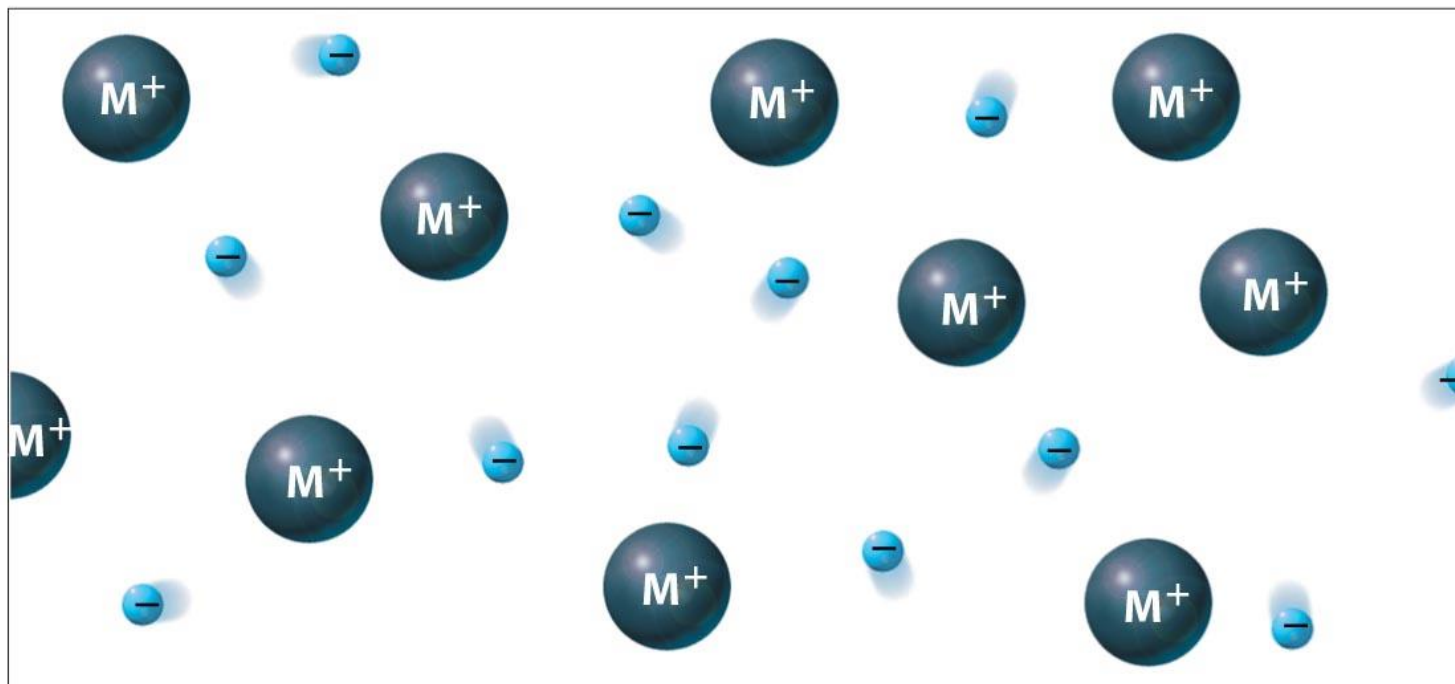
Nonpolar Covalent bond

WARM-UP

Draw a Lewis dot diagram for each:

- CO_2
- Na_2S
- BF_3 (remember the exception for B?)
- NH_3
- CH_4

2:00



**METALLIC BONDING — “(+) IONS
IN A SEA OF E^- ”**

Occurs between **METALS**

**Attractions between
metal cations and
delocalized valence
electrons**

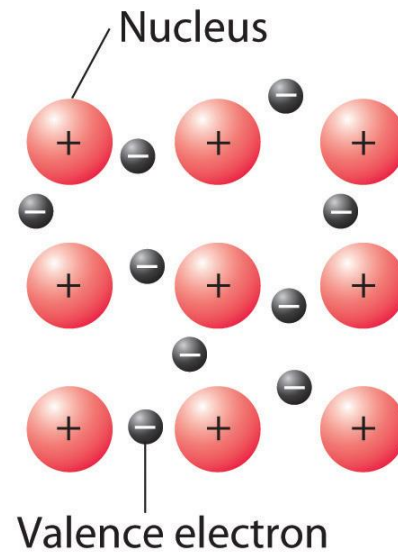
The bond forms from
metal atoms losing their
valence electrons. These
electrons are then free to
move throughout the
metallic crystal and act as
the “glue” that holds the
metal ions together

STRENGTH OF THE METALLIC BOND IS DETERMINED BY:

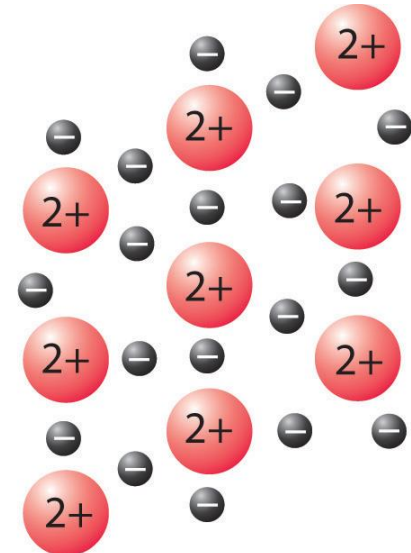
Number of delocalized electrons

The greater number of delocalized valence electrons = more “glue” to hold the metal cations in the metallic crystal together = stronger metal

Hardest metals = transition metals



(a) Group 1 metal



(b) Group 2 metal

PROPERTIES OF METALLIC CRYSTALS

Hard, metallic crystals

Good conductors of heat and electricity

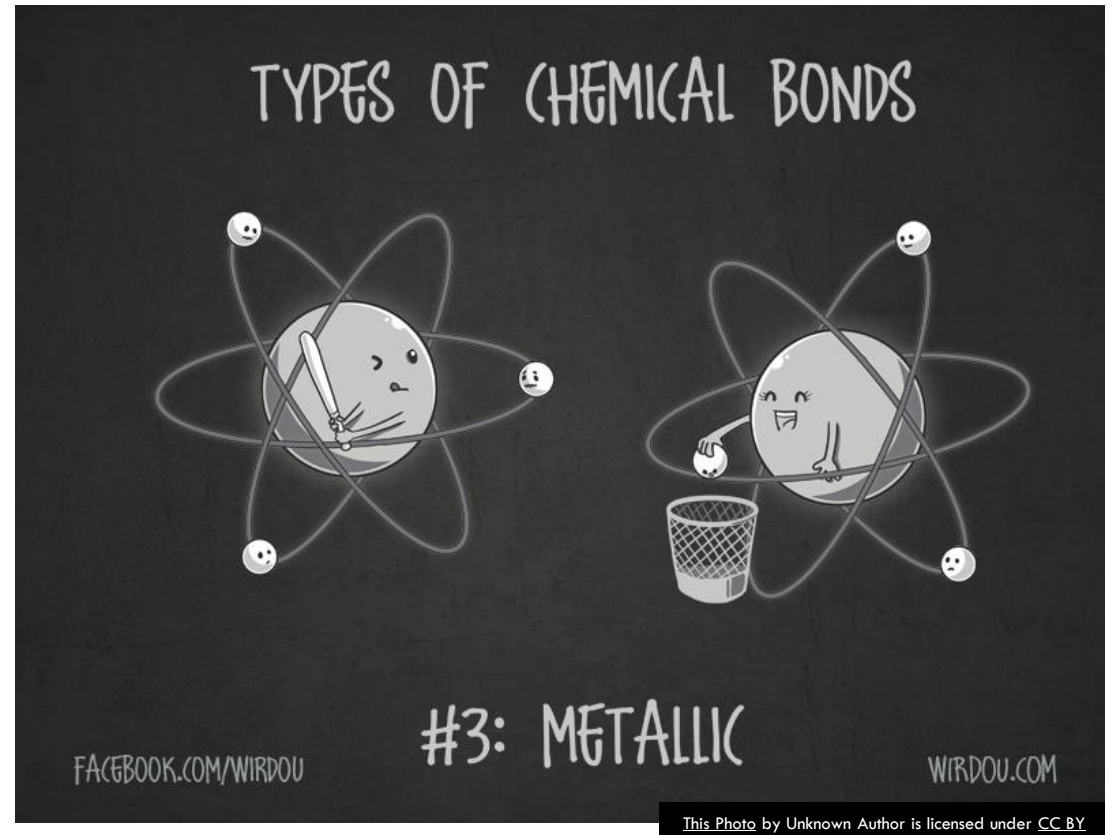
- Because e^- are free to move

High melting and boiling points

- Strong bond that requires a lot of energy to break

Shiny (luster)

- Because e^- interact with light (photoelectric effect)



PROPERTIES OF METALLIC CRYSTALS

Malleable (hammered in thin sheets)

Ductile (drawn into thin wires)

- Because e⁻ can move

Insoluble in water

- Because it is a strong bond that has a stronger attraction to itself than to water



WARM-UP

What are the four types of covalent bonding?

- Describe each AND
- Give an example of each

MOLECULAR GEOMETRY = MOLECULAR SHAPE

VSEPR Theory (Valence Shell Electron Pair Repulsion)

- e- pairs (clouds) spread as far apart as possible to minimize repulsive forces
- Unshared pairs take up more space than shared pairs

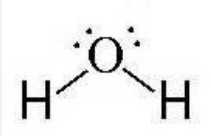
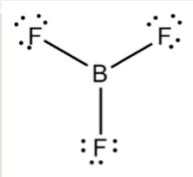
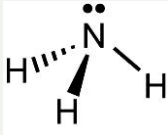
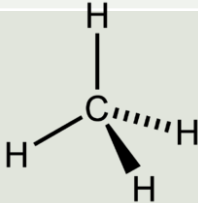
WARM-UP

Ionic or covalent?

- SrO
- PCl₃

Draw Lewis dot structures for the two compounds above.

BASIC MOLECULAR GEOMETRIES

Shape	# Bonds to Central Atom	# Lone Pairs to Central Atom	Examples	Bond Angles
Linear	2 atoms together or 2 bonds to central atom	0	H-Cl $\text{O}=\text{C}=\text{O}$	180°
Bent	2	2		104.5°
Trigonal Planar	3	0		120°
Trigonal Pyramidal	3	1		107.5°
Tetrahedral	4	0		109.5°

MOLECULAR POLARITY

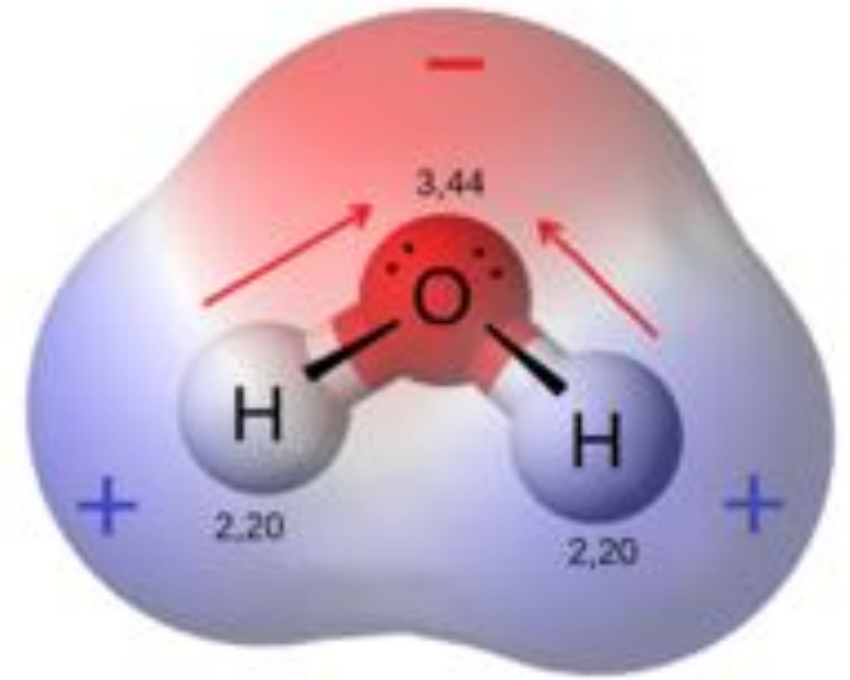
Molecular polarity: distribution of molecular charge (even or uneven)

Molecular polarity depends on:

- Bond polarity – even or uneven distribution of valence electrons in the bond between two atoms
- **Molecular shape – symmetrical or not**

Molecular polarity influences:

- Intermolecular forces



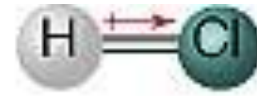
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HOW TO DETERMINE POLARITY OF MOLECULES

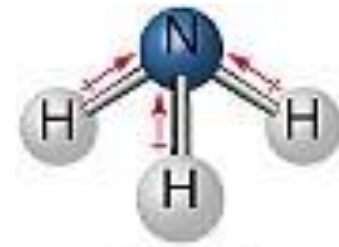
Nonpolar molecules are
Symmetrical

Polar molecules are
Asymmetrical

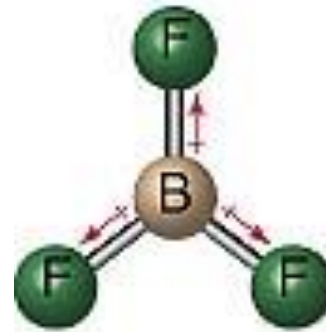
Polar molecules are called
dipoles (have (-) and (+)
ends of the molecules)



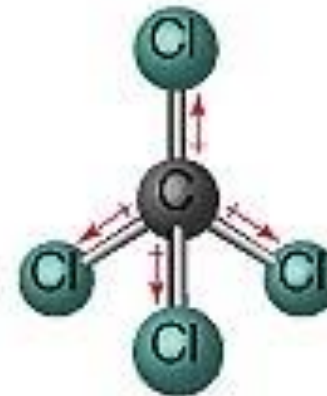
Polar



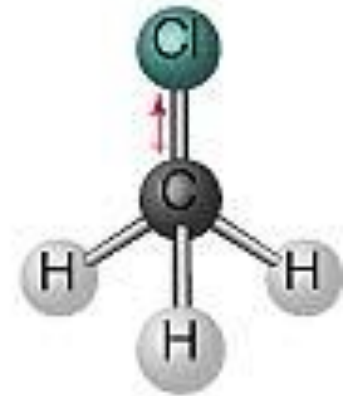
Polar



Nonpolar



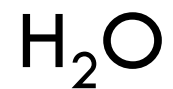
Nonpolar



Polar

LET'S TRY IT!

Name and sketch the shape of the following. Is it polar or nonpolar?



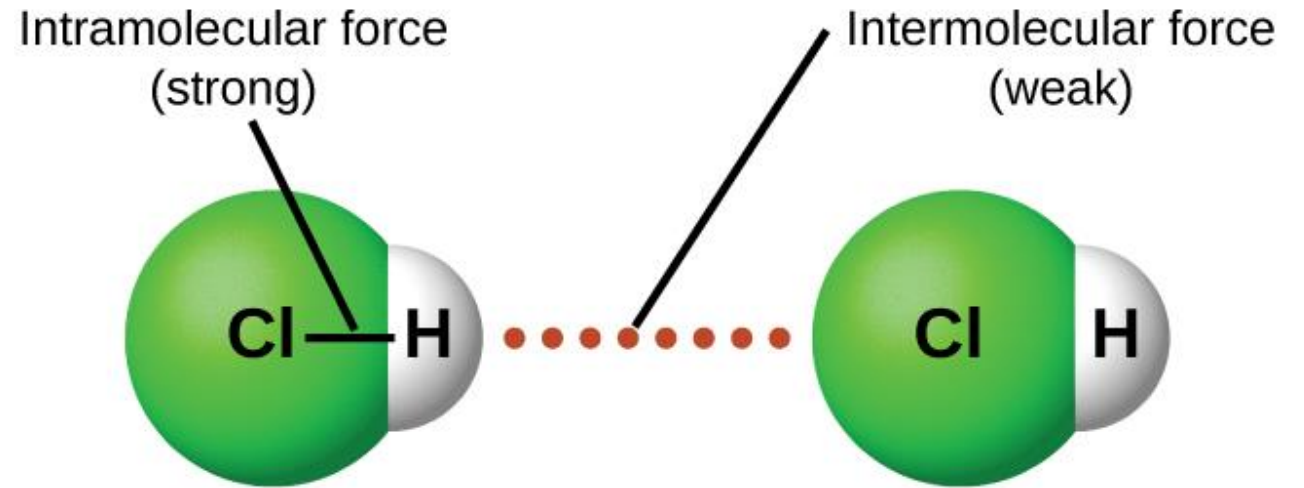
WARM-UP

For each molecule, draw the structure, identify its shape and indicate if the molecule is polar or nonpolar

- CHCl_3
- CO_2
- CH_3OH

Molecular Shape	Polar	Nonpolar
Linear	High diff in electronegativity	Bonded atoms are the same (diatomic), or small diff in electronegativity
Bent	Always Polar	
Trigonal Planar	Bonded atoms are different elements	All bonded atoms are the same element
Tetrahedral	Bonded atoms are different elements	All bonded atoms are the same element
Trigonal Pyramidal	Always Polar	

INTERMOLECULAR FORCES



Definition: weak forces of attraction between molecules
(van der Waals forces)

There are 4 types of intermolecular forces (imfs)

- Dipole-Dipole forces
- Dipole-Induced Dipole forces
- Hydrogen Bonding
- London Dispersion forces

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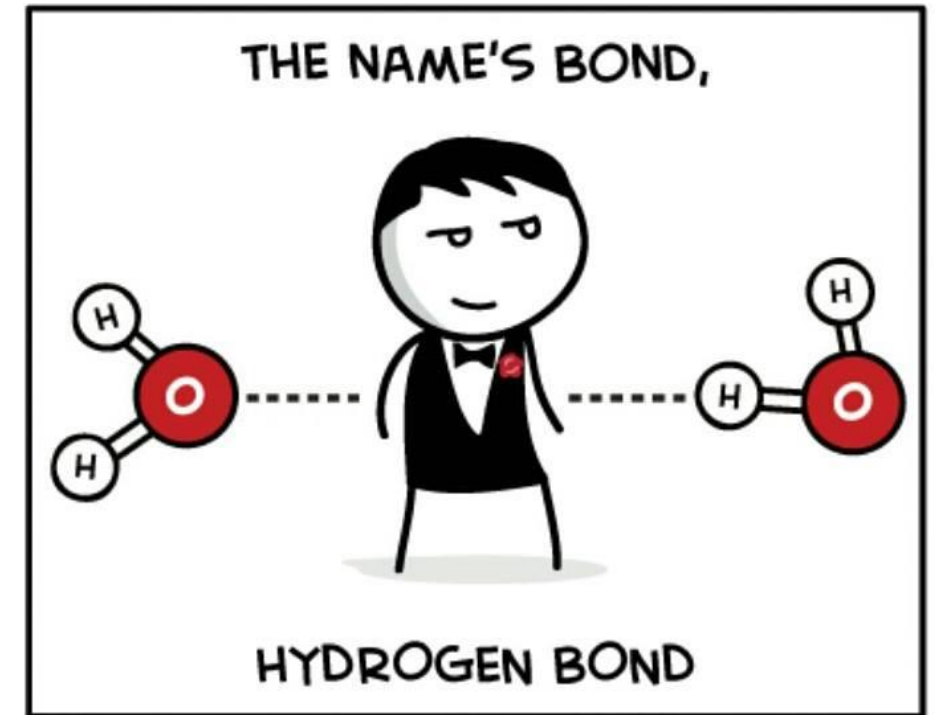
HYDROGEN BONDING

This is a special dipole-dipole force that is the strongest of the intermolecular forces

Occurs when H of one molecule is attracted to N, O, or F of another molecule

This imf gives water its unusual properties:

- Ice floating
- Higher melting and boiling points
- Surface tension



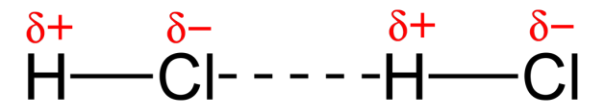
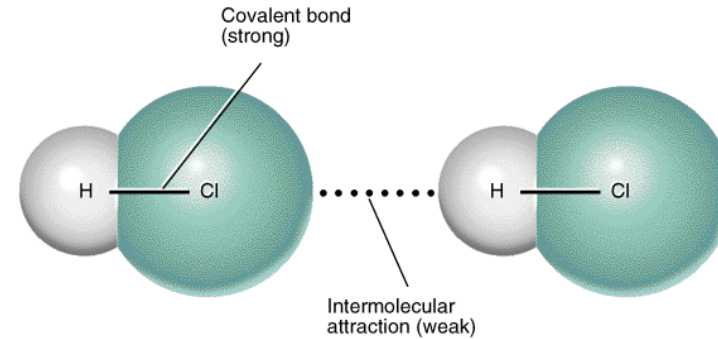
VICTIMS OF CIRCUMSOLAR

DIPOLE-DIPOLE FORCES

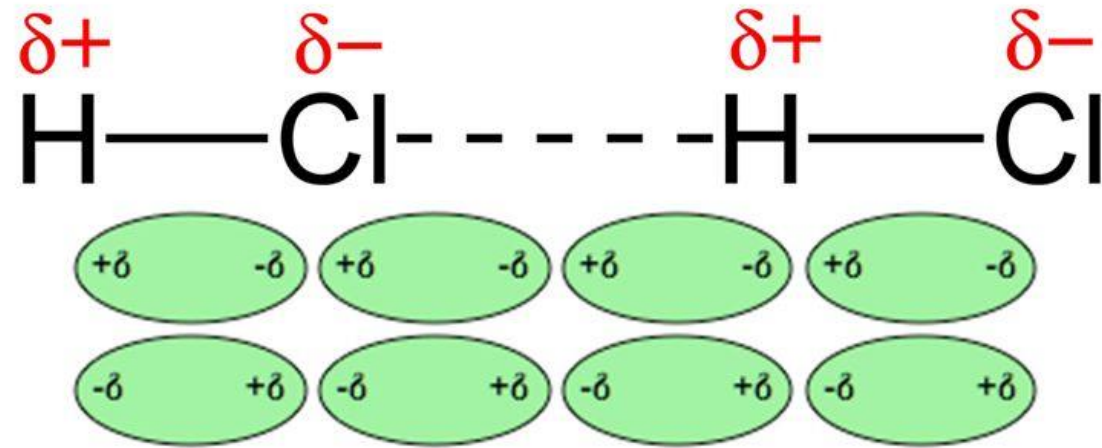
Exist between polar molecules

Causes molecules to have higher melting points and boiling points than expected

Substances exist mostly as solids or liquids due to the strength of the imf



Dipole-Dipole Forces



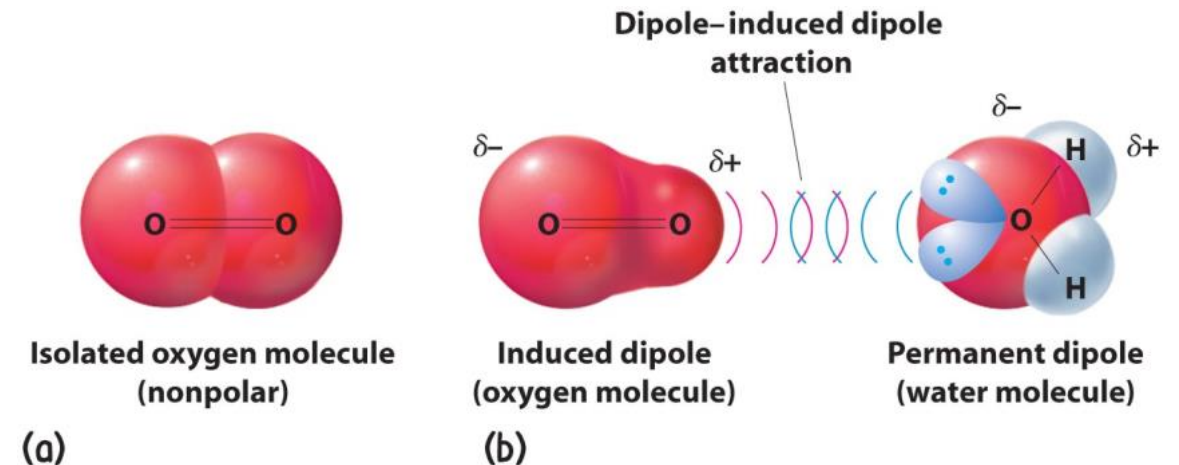
- The + and – ends of polar molecules attract one another.
- Dipole-dipole interactions are *similar to* but *much weaker* than ionic bonds.

DIPOLE-INDUCED DIPOLE FORCES

Exists between polar and nonpolar molecules

Created by a temporary shift in e^- of the nonpolar molecule due to the presence of the polar molecule or dipole

Ex. O_2 (NP) can dissolve in H_2O (P)



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LONDON DISPERSION FORCES

This is the **ONLY** type of imf that can occur in **NONPOLAR** molecules

Creates an instantaneous dipole due to the shift in e^- of a neighboring nonpolar molecule

This is the weakest imf and its strength increases with increasing number of electrons

LONDON DISPERSION FORCES

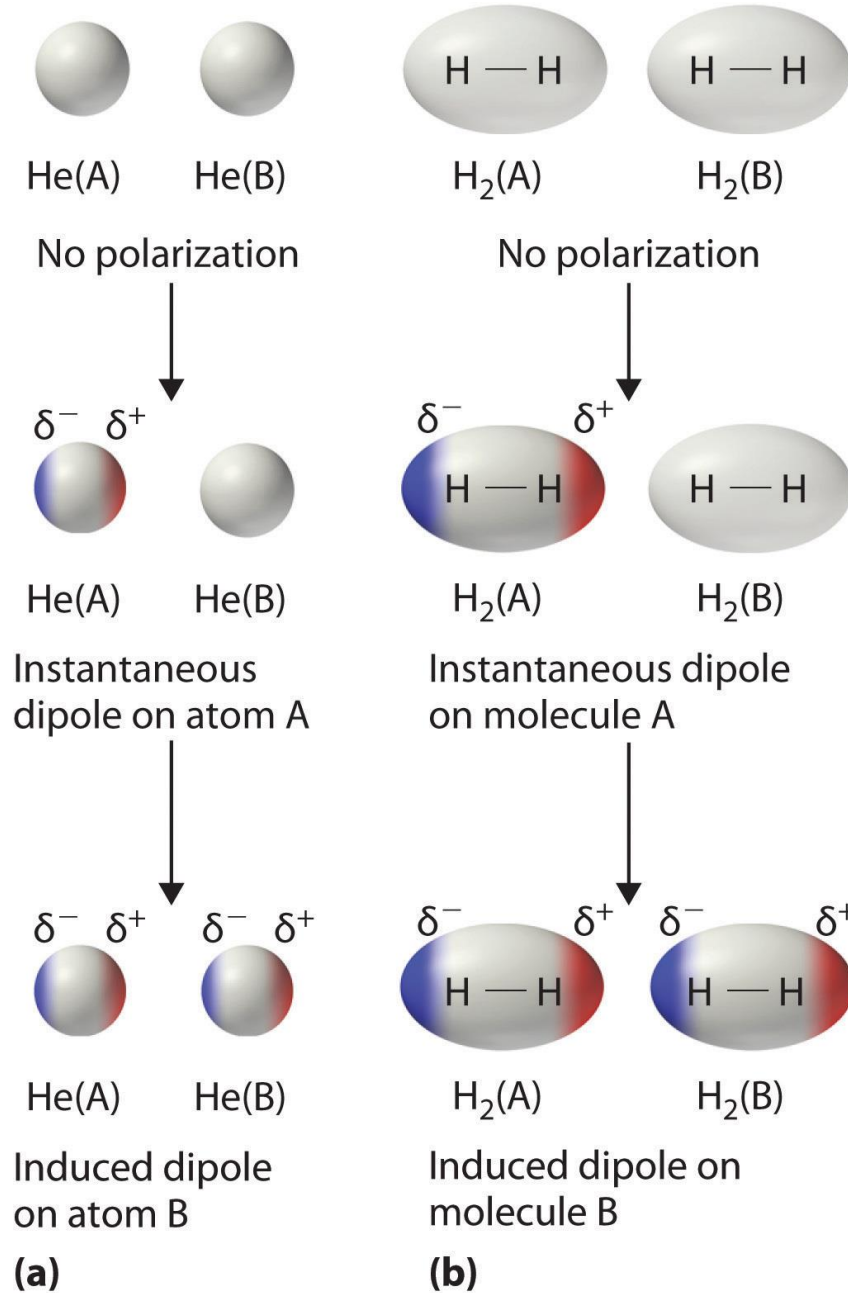
This is the weakest imf and its strength increases with increasing number of electrons

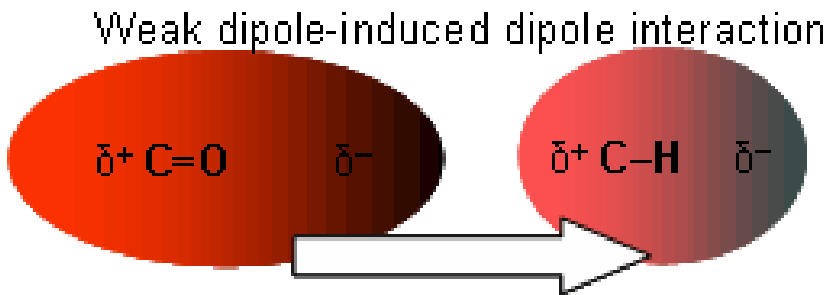
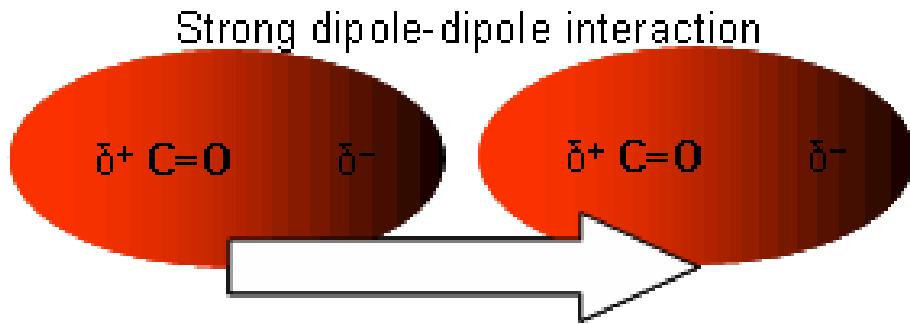
Causes molecules to have low melting and boiling points

Most of the substances with only LDF imfs are gases

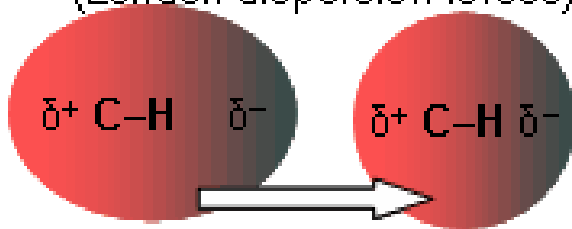
- Ex. “HOFBrINCl”, noble gases

Special Note: This imf exists between all molecules as it is dependent on the number of electrons in a molecule





Very weak transient dipole interactions
(London dispersion forces)



Ranking the Intermolecular Forces:

Strongest to Weakest

Hydrogen Bonding

Dipole-Dipole Forces

Dipole-Induced Dipole Forces

London Dispersion Forces

PHYSICAL PROPERTIES AND BONDING

Melting Point – temperature at which solid $\rightarrow$ liquid

Boiling Point – temperature at which liquid $\rightarrow$ gas

Density = mass/volume (g/mL or g/cm³)

Color – some transition metals produce colored ions in solution

- Ex. Cu – blue/green
- Metals are shiny – e- interact with light

PHYSICAL PROPERTIES AND BONDING

Color – some transition metals produce colored ions in solution

- Ex. Cu – blue/green
- Metals are shiny – e- interact with light

Solubility – ability of a solute to dissolve in a given amount of solvent
→ solution

- “Like dissolve like”

HOW DO THESE PROPERTIES RELATE TO BOND TYPE OR IMF?

Stronger bonds/imfs $\rightarrow$ higher melting/boiling points

Weaker bonds/imfs $\rightarrow$ lower melting/boiling points

WARM-UP

Polar or nonpolar? What is the strongest IMF for each?

- SBr_2
- CS_2
- HF
- H_2O with Cl_2

WARM-UP

List the following in order of increasing atomic radius: V, Ga, K, Br

Draw the following Lewis dot structures:

- Na_3N
- SF_2

TEST TOPICS

Review all vocab!

Unit 2 Stuff – Periodic Table and Quantum

Ionic Bonding

- Determining Formula
- Lewis dot structures
- With PAI's too!

Metallic Bonding

Covalent Bonding – All four types!

- Lewis dot structures
- VSEPR Shape
- Polarity
- IMF's

Properties of each bond type